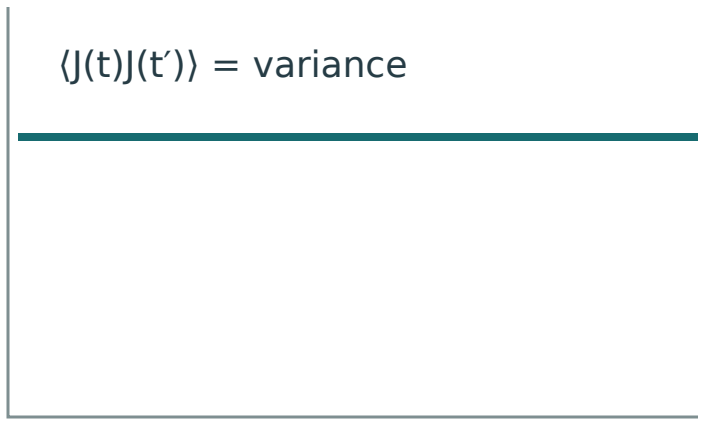
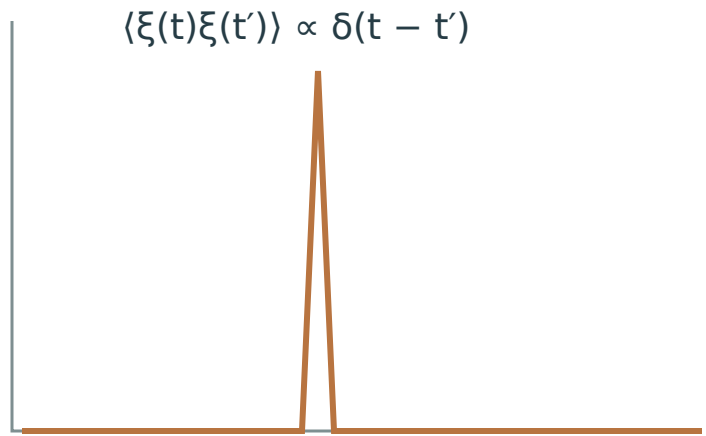
The same spatial randomness can have different memory

Quenched coupling J



After averaging: two independent time integrals

Markov noise ξ(t)



After averaging: a local time integral

Schematic kernels, not spectral data. The delta peak is drawn with finite width.